

Hochschild Homology of the $\mathfrak{b}$ -Pseudodifferential Symbol Algebra

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Abstract

Let X be a compact n -manifold with embedded boundary and let $E \rightarrow X$ be a complex vector bundle. We compute the continuous Hochschild homology of the Benameur–Nistor Laurent complete-symbol algebra of the b-groupoid:

$$\mathcal{A}_{b,L}(X;E) = \mathcal{O}(X) \left(\Psi^\infty(\mathcal{G}_b;E) / \Psi^{-\infty}(\mathcal{G}_b;E) \right).$$

Here $\mathcal{O}(X)$ consists of functions with finite Laurent singularities at the boundary, and the algebra carries the published topologically filtered completion. With completed continuous Hochschild chains,

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}({}^bS^*X \times S_\sigma^1; \mathbb{C}).$$

The circle S_σ^1 records symbol order and H_L^* is Laurent de Rham cohomology at the boundary. The displayed vector-space isomorphism is noncanonical; the filtration and its associated graded are natural. We also give two alternative reconstructions: a Čech–de Rham descent proof of the filtered calculation and a Rees–Fedosov reconstruction based on Lie-algebroid Hochschild-chain formality. In the second route the formal-to-classical comparison is made at every finite symbol order, while convergence for the actual continuous algebra remains the quoted Benameur–Nistor input. The Rees parameter is not confused with an additional geometric circle. We separate this result from the Hochschild homology of an unreduced operator algebra, which depends on the chosen residual ideal and completion.

1 The precise statement

The phrase *the algebra generated by b-pseudodifferential operators* is ambiguous. At least four choices affect Hochschild homology: the allowed operator orders, the allowed powers of a boundary defining function, the residual ideal, and the locally convex completion. We use the standard algebra for which the homology has a finite-dimensional geometric description. It is the b-groupoid specialization of the Laurent symbol algebras studied in [8, 2, 1].

For one boundary hypersurface, the standard b-groupoid has underlying set

$$\mathcal{G}_b = X^\circ \times X^\circ \sqcup \bigsqcup_{H \in \pi_0(\partial X)} H \times H \times \mathbb{R}.$$

On a boundary component, composition adds the last coordinate. Its smooth structure is fixed by the normal ratio $t = \log(x/x')$, and its Lie algebroid is $A(\mathcal{G}_b) = {}^bTX$. For corners, one uses the standard groupoid with an $\mathbb{R}^j$ -factor over each open codimension- j face [1].

Let $\mathcal{O}(X)$ be the ring generated by $C^\infty(X)$ and x_H^{-1} for all boundary hypersurfaces H . Benameur and Nistor define

$$\mathcal{A}_{b,L}(X;E) := \mathcal{O}(X) \left(\Psi^\infty(\mathcal{G}_b; r^*E) / \Psi^{-\infty}(\mathcal{G}_b; r^*E) \right)$$

with their topologically filtered topology and completed Hochschild complex [2, 1]. The vector representation identifies this with the usual properly supported Laurent b-complete symbols. In particular, one may write its underlying filtered union heuristically as

$$\frac{\bigcup_{m,\ell} x^{-\ell} \Psi_b^m(X;E)}{\bigcup_{\ell} x^{-\ell} \Psi_b^{-\infty}(X;E)},$$

but this quotient notation does not by itself specify the topology used in the theorem.

Changing the defining function from x to $x' = e^f x$ gives

$$(x')^{-\ell} = e^{-\ell f} x^{-\ell}.$$

Hence the change only multiplies a Laurent generator by a smooth invertible function, and the algebra, its filtration, and the answer below are canonically independent of this choice. For corners one allows an arbitrary finite Laurent monomial $\prod_H x_H^{-\ell_H}$, one factor for each boundary hypersurface.

Set $Y = {}^b S^* X = ({}^b T^* X \setminus 0) / \mathbb{R}_+$ and $Z = Y|_{\partial X}$. Thus $\partial Y = Z$. The symbol-order circle is denoted S_σ^1 .

Theorem 1.1 (Hochschild homology). *For every $k \geq 0$,*

$$\boxed{\mathrm{gr} \, \mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X; E)) \cong H_L^{2n-k}(Y \times S_\sigma^1; \mathbb{C}).} \quad (1)$$

After choosing splittings of the finite homology filtration, this gives a noncanonical vector-space isomorphism

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X; E)) \cong H_L^{2n-k}(Y \times S_\sigma^1; \mathbb{C}).$$

The answer is independent of E . If X has one embedded boundary hypersurface, then

$$\begin{aligned} \mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X; E)) &\cong H^{2n-k}(Y) \oplus H^{2n-k-1}(Y) \\ &\quad \oplus H^{2n-k-1}(Z) \oplus H^{2n-k-2}(Z), \end{aligned} \quad (2)$$

where a cohomology group in negative degree is zero. In particular the homology vanishes for $k > 2n$.

The associated-graded statement in (1) is intrinsic. The four-summand formula uses choices of Laurent face splitting and the Künneth splitting for S_σ^1 .

$$\begin{array}{ccccc} \mathcal{A}_{b,L} & \xrightarrow{\text{order filtration}} & \mathrm{gr} \, \mathcal{A}_{b,L} & \xrightarrow{\mathrm{HKR}} & \Omega_L^*({}^b T^* X \setminus 0) \\ & & & & \downarrow \delta_{\pi_b} \\ \mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L}) & \xleftarrow{E^2=E^\infty} & H_L^{2n-*}(Y \times S_\sigma^1) & \xleftarrow{\text{symplectic duality}} & H_*^{\pi_b, L}({}^b T^* X \setminus 0) \end{array}$$

Figure 1: The computation passes from the filtered noncommutative algebra to its commutative associated graded, then from Poisson homology to Laurent de Rham cohomology.

2 Continuous chains and the two filtrations

The topology is part of the theorem. For a topologically filtered algebra A , Benamèur–Nistor do *not* take the completed projective tensor power of the whole locally convex space: multiplication need not be jointly continuous for that topology. In their notation the degree- k chain space is

$$H_k(A) = \varinjlim_N (F_{N,N}^N A)^{\widehat{\otimes}_{\pi}(k+1)}. \quad (3)$$

Here $F_{N,N}^N A$ is the diagonal cofinal family of filtration pieces from [1, Definition 1]; equivalently, one first takes the projective completion in the decreasing symbol-order direction and only then enlarges the support and Laurent bounds by an inductive limit. We set

$$C_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) := H_k(\mathcal{A}_{b,L}).$$

Whenever a tensor product below carries no decoration it denotes an elementary tensor in the dense filtered subcomplex of (3), not an ordinary completed tensor power of the whole algebra.

For a chain $a_0 \otimes \cdots \otimes a_k$, the Hochschild boundary is

$$b(a_0 \otimes \cdots \otimes a_k) = \sum_{j=0}^{k-1} (-1)^j a_0 \otimes \cdots \otimes a_j a_{j+1} \otimes \cdots \otimes a_k \\ + (-1)^k a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}.$$

Multiplication is continuous on fixed filtration pieces, so b extends to the completed complex and satisfies $b^2 = 0$.

Laurent order and symbol order play different roles. The first is built into $\mathcal{O}(X)$; symbol order produces the spectral sequence. On every fixed piece in (3), filter elementary tensors by

$$\text{ord}(a_0) + \cdots + \text{ord}(a_k) \leq p,$$

take the projective closure there, and then pass through the same projective/inductive limits. Denote the resulting subspace by $\mathcal{F}_p C_k$. Then $b(\mathcal{F}_p C_k) \subseteq \mathcal{F}_p C_{k-1}$. Exhaustivity, completeness, and the convergence hypotheses are those verified in the Benaméur–Nistor topologically filtered theorem. Borel quantization explains asymptotic completeness of symbols, but it is not by itself a convergence proof for an arbitrarily completed Hochschild complex.

Discussion. Why divide by smoothing operators? The principal-symbol filtration sees an operator only through its full asymptotic symbol. Smoothing kernels form the kernel of that passage. Keeping them gives an extension problem, treated separately in Section 12; it is not a harmless change of notation.

3 The b-cotangent geometry

The Lie algebra of b-vector fields is

$$\mathcal{V}_b(X) = \{V \in C^\infty(X; TX) : V \text{ is tangent to } \partial X\}.$$

It is the space of smooth sections of a vector bundle ${}^bTX \rightarrow X$. Near a boundary point, choose product coordinates

$$(x, y_1, \dots, y_{n-1}).$$

Then

$$\mathcal{V}_b(X) = C^\infty(X)\text{-span}\{x\partial_x, \partial_{y_1}, \dots, \partial_{y_{n-1}}\}.$$

The dual local frame of ${}^bT^*X$ is

$$\frac{dx}{x}, \quad dy_1, \dots, dy_{n-1}.$$

Thus a b-covector has the form

$$\xi \frac{dx}{x} + \eta \cdot dy.$$

On ${}^bT^*X$ the tautological b-one-form and its differential are

$$\alpha_b = \xi \frac{dx}{x} + \eta \cdot dy, \quad \omega_b = d\alpha_b = d\xi \wedge \frac{dx}{x} + d\eta \wedge dy.$$

The form ω_b is nondegenerate as a b-two-form. Its inverse is the b-Poisson tensor

$$\pi_b = x\partial_x \wedge \partial_{\xi} + \sum_j \partial_{y_j} \wedge \partial_{\eta_j}.$$

This is the geometric bracket appearing in the first nonzero differential of the Hochschild spectral sequence.

Proposition 3.1 (Anchor bridge). *The anchor $a : {}^bTX \rightarrow TX$ becomes an isomorphism after adjoining Laurent coefficients:*

$$\mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma({}^bTX) \xrightarrow{a} \mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma(TX).$$

After dualizing, this is not a smooth vector-bundle isomorphism at the boundary. It is an isomorphism only after Laurent localization:

$$\mathcal{O}(X) \otimes C^\infty({}^bT^*X) \cong \mathcal{O}(X) \otimes C^\infty(T^*X)$$

as fiberwise-polynomial Poisson algebras, extended by completion to the homogeneous Laurent symbol complexes. Consequently the anchor lemmas of [1] give a chain-level isomorphism of the corresponding homogeneous Laurent de Rham complexes and hence

$$H_L^*({}^bS^*X \times S_\sigma^1) \cong H_L^*(S^*X \times S_\sigma^1).$$

Proof. Locally the anchor sends $x\partial_x$ to the ordinary vector field $x\partial_x$ and fixes ∂_{y_j} . Its localized inverse uses the Laurent coefficient x^{-1} : $\partial_x \mapsto x^{-1}(x\partial_x)$. The map preserves brackets, so its localized dual preserves the linear Poisson brackets. Taking the degree-zero homogeneous subcomplex and adjoining the order-circle generator therefore gives the stated cohomology isomorphism. This is the one-fibre specialization of the anchor lemmas in [1]; no isomorphism ${}^bT^*X \cong T^*X$ of smooth bundles is being asserted. $\square$

4 The associated graded algebra

Filter $\mathcal{A}_{b,L}$ by operator order:

$$F_m \mathcal{A}_{b,L} = \text{image} \left(\bigcup_{\ell \geq 0} x^{-\ell} \Psi_b^m(X; E) \right), \quad m \in \mathbb{Z}.$$

The filtration is exhaustive, separated in the complete-symbol sense, and multiplicative. Principal symbols give

$$\text{gr}_m \mathcal{A}_{b,L} \cong \mathcal{O}_L({}^bT^*X \setminus 0; \text{End } E)_m,$$

the Laurent functions that are homogeneous of degree m in the fibers. In a b-Darboux chart, a choice of local Weyl quantization gives the expansion

$$a \# c \sim ac + \frac{1}{2i} \{a, c\}_b + \text{terms two orders lower}, \quad (4)$$

where changing quantization changes the individual lower-order symmetric terms. Without a chosen global symplectic connection, the full Weyl formula is not coordinate invariant. What is invariant, and all that is used below, is the commutator symbol

$$\sigma_{m+m'-1}([A, C]) = i^{-1} \{a, c\}_b.$$

This one-order drop is the origin of the first nonzero differential.

Lemma 4.1 (Morita reduction). *The coefficient bundle E does not change the Hochschild homology.*

Proof. The global module of complete symbols with coefficients in E is a finitely generated projective equivalence bimodule between the scalar algebra and its End E -coefficient algebra. The associated matrix-trace and corner maps are continuous and filtration preserving, and the standard matrix-unit homotopy proves Morita invariance. This is a global Morita equivalence; no partition of unity is used to glue local chain homotopies. The topologically filtered version, including graded bundles, is the coefficient-independence lemma of [2, 1]. $\square$

We therefore suppress E . The Hochschild–Kostant–Rosenberg map on the associated graded algebra is

$$\chi(a_0 \otimes \cdots \otimes a_j) = \frac{1}{j!} a_0 da_1 \wedge \cdots \wedge da_j.$$

Proposition 4.2 (The E^1 -page). *The HKR map identifies the first page of the order spectral sequence with Laurent differential forms on ${}^bT^*X \setminus 0$, decomposed by fiber homogeneity.*

Proof. On each b-cotangent chart the associated graded product is ordinary multiplication of scalar homogeneous symbols. Antisymmetrization gives the inverse to χ on Hochschild homology:

$$a_0, da_1 \wedge \cdots \wedge da_j \longmapsto \sum_{\tau \in S_j} \text{sgn}(\tau) a_0 \otimes a_{\tau(1)} \otimes \cdots \otimes a_{\tau(j)}.$$

The displayed antisymmetrization is not a literal chain-level inverse; the two composites become inverse after the Koszul homotopy. For the topologically filtered Laurent groupoid algebra, continuity, completion, and globalization are the HKR proposition used by Benamèur–Nistor [2, 1]. It may be proved through the sheaf of local Koszul resolutions and its fine resolution, but a bare partition of unity does not patch chain homotopies. Fiber dilation commutes with the published quasi-isomorphism, so it retains homogeneity and gives

$$E_{p,q}^1 \cong \Omega_L^{p+q}({}^bT^*X \setminus 0)_p,$$

with the conventional reindexing determined by total symbol order. $\square$

5 The first differential is Poisson homology

For classical b-symbols a and c , symbolic composition gives

$$\sigma_{m+m'-1}([A, C]) = \frac{1}{i} \{a, c\}_b.$$

Consequently the first differential after HKR is, up to the harmless factor i , the Poisson differential

$$\delta_b = [\iota_{\pi_b}, d] = \iota_{\pi_b} d - d \iota_{\pi_b}.$$

Explicitly,

$$\begin{aligned} \delta_b(a_0, da_1 \wedge \cdots \wedge da_j) &= \sum_{r=1}^j (-1)^{r+1} \{a_0, a_r\}_b da_1 \wedge \cdots \widehat{da_r} \cdots \wedge da_j \\ &\quad + \sum_{1 \leq r < s \leq j} (-1)^{r+s} a_0 d\{a_r, a_s\}_b \wedge da_1 \wedge \cdots \widehat{da_r} \cdots \widehat{da_s} \cdots \wedge da_j. \end{aligned} \quad (5)$$

To see this differential directly, insert the order-one-lower part of (4) into each multiplication in the Hochschild boundary. The symmetric terms cancel after HKR antisymmetrization, while the commutator terms combine to (5).

Let $N = 2n$ be the dimension of ${}^bT^*X$, put $\mu_b = \omega_b^n/n!$, and let $\pi_b^\sharp : \Omega^j \rightarrow \Lambda^j T$ be contraction of every covector with π_b . Symplectic duality is the map

$$*_b : \Omega_L^j({}^bT^*X \setminus 0) \longrightarrow \Omega_L^{N-j}({}^bT^*X \setminus 0), \quad *_b \alpha = \iota_{\pi_b^\sharp \alpha} \mu_b,$$

and $*_b^2 = 1$ up to the standard degree sign. Applying the two sides to a monomial in the Darboux b-frame

$$\frac{dx}{x}, dy_1, \dots, dy_{n-1}, d\zeta, d\eta_1, \dots, d\eta_{n-1}$$

gives, with the displayed convention for δ_b ,

$$*_b \delta_b = (-1)^{j+1} d *_b \quad \text{on } \Omega_L^j. \quad (6)$$

The calculation remains valid at $x = 0$: the displayed frame is smooth as a b-frame, and both operators preserve finite Laurent pole order. Therefore a δ_b -cycle of degree j is carried to a closed Laurent form of degree $2n - j$, while boundaries are carried to exact forms.

Corollary 5.1. *The Poisson homology on the E^2 -page is Laurent de Rham cohomology with complementary degree:*

$$H_j^{\pi_b, L}({}^bT^*X \setminus 0) \cong H_L^{2n-j}({}^bT^*X \setminus 0).$$

6 Homogeneity and the symbol-order circle

Choose a positive fiber norm r on ${}^bT^*X \setminus 0$. Then

$${}^bT^*X \setminus 0 \cong Y \times \mathbb{R}_+, \quad (q, r) \longmapsto rq.$$

The order index in the spectral sequence and the degree reversal under $*_b$ combine so that its relevant de Rham complex is the total homogeneity-zero complex. This bookkeeping matters: δ_b lowers fiber homogeneity by one, whereas $*_b$ changes both form degree and homogeneity. After the complementary-degree regrading, the surviving weight is zero, as in the complete-symbol theorem of Brylinski–Getzler and Benaneur–Nistor.

The zero-weight reduction itself is elementary. If $\mathcal{E} = r\partial_r$ is the Euler vector field and a form α has nonzero homogeneity w , Cartan’s formula gives

$$(d\iota_{\mathcal{E}} + \iota_{\mathcal{E}}d)\alpha = \mathcal{L}_{\mathcal{E}}\alpha = w\alpha.$$

Thus $w^{-1}\iota_{\mathcal{E}}$ contracts every nonzero weight.

At weight zero, restriction to $r = 1$ gives a unique splitting

$$\alpha(r, q) = \alpha_0(q) + \frac{dr}{r} \wedge \alpha_1(q).$$

Since $d(dr/r) = 0$, its differential is

$$d\alpha = d_Y \alpha_0 - \frac{dr}{r} \wedge d_Y \alpha_1.$$

Thus the weight-zero complex is the direct sum of the de Rham complex of Y and a copy shifted by one degree. The complete-symbol order is an integer grading with no preferred logarithm; cohomologically this two-term complex is canonically written as the de Rham complex of an auxiliary circle:

$$\alpha_0 + \frac{dr}{r} \wedge \alpha_1 \longleftrightarrow \alpha_0 + d\theta_\sigma \wedge \alpha_1.$$

The formal order variable dr/r is represented cohomologically by the generator $d\theta_\sigma$ of a circle. Hence

$$H_L^j({}^bT^*X \setminus 0)_{\text{hom}=0} \cong H_L^j(Y \times S_\sigma^1). \quad (7)$$

This is the source of the extra S^1 in the answer; it is not a geometric circle in the fibers of ${}^bT^*X$, and no choice of fiber norm affects its cohomology class.

7 Laurent cohomology and the boundary face

Let M be a compact manifold with one embedded boundary hypersurface and defining function x . The Laurent de Rham complex is

$$\Omega_L^j(M) = \bigcup_{\ell \geq 0} x^{-\ell} \Omega^j(M), \quad d : \Omega_L^j(M) \rightarrow \Omega_L^{j+1}(M).$$

Proposition 7.1 (Face decomposition). *There is an isomorphism*

$$H_L^j(M) \cong H^j(M) \oplus H^{j-1}(\partial M).$$

The boundary summand is represented by $\tilde{\beta} \wedge d(\chi \log x)$, where β is a closed $(j-1)$ -form on ∂M , $\tilde{\beta}$ is a collar extension, and χ is a cutoff supported in the collar and equal to one near the boundary.

Proof. Work in a collar and write a Laurent form uniquely as

$$\alpha(x) + \frac{dx}{x} \wedge \beta(x),$$

where the tangential forms $\alpha(x), \beta(x)$ are Laurent families with a finite principal part and a smooth Taylor tail. Filter the collar complex by pole order and then by normal jet. On its associated graded, a term has a definite normal weight q . For $q \neq 0$, Cartan's identity gives the contracting homotopy

$$h_q = q^{-1} \iota_{x\partial_x}, \quad dh_q + h_q d = 1.$$

Induction over the finite negative weights removes the principal part; the remaining positive Taylor tail is smooth and belongs to the ordinary de Rham complex. This jet-filtration argument, rather than freezing all coefficients at $x = 0$, is the Laurent Poincaré lemma. The only new normal class is the weight-zero generator dx/x .

In particular, after writing

$$\alpha = \alpha_0 + \frac{dx}{x} \wedge \beta_0 + \text{terms of nonzero normal weight},$$

the residue is

$$\text{Res}_{\partial M} : \Omega_L^j(M) \longrightarrow \Omega^{j-1}(\partial M), \quad \text{Res}_{\partial M}(\alpha) = \beta_0|_{\partial M}.$$

A direct calculation in the displayed order gives

$$\text{Res}(d\alpha) = -d_{\partial M} \text{Res}(\alpha).$$

Thus residue is a chain map to the degree-shifted boundary complex. The Laurent Poincaré lemma identifies its homotopy kernel with $\Omega^*(M)$, yielding the quasi-isomorphism

$$\Omega_L^*(M) \simeq \Omega^*(M) \oplus \Omega^{*-1}(\partial M).$$

If β is closed of degree $j-1$, let $p : [0, \varepsilon)_x \times \partial M \rightarrow \partial M$ be the collar projection. Choose χ with support in this collar and choose $\tilde{\beta} = p^*\beta$ on an open set containing $\text{supp } d(\chi \log x)$. Extend it arbitrarily beyond that set. Then $d\tilde{\beta} = 0$ everywhere that $d(\chi \log x)$ is nonzero, so the representative below is closed. This collar construction gives the cohomological splitting

$$\beta \longmapsto (-1)^{j-1} \tilde{\beta} \wedge d(\chi \log x).$$

Its residue is β . If $x' = e^f x$, then $d \log x' = d \log x + df$; changing the defining function, collar, or cutoff changes this splitting by an ordinary/exact correction. Hence the direct sum is natural as a filtered extension but its explicit vector-space splitting is not canonical. $\square$

Discussion. Where do higher poles go? A pole such as $x^{-q-1}dx$, $q > 0$, looks more singular than dx/x , but it is $-q^{-1}d(x^{-q})$. Higher Laurent coefficients occur in the complex and are essential for closure under composition; they add no new normal cohomology. Only the logarithmic coefficient survives.

Apply Proposition 7.1 to $M = Y \times S_\sigma^1$, whose boundary is $Z \times S_\sigma^1$. Then use

$$H^r(W \times S^1) \cong H^r(W) \oplus H^{r-1}(W).$$

For $r = 2n - k$ this gives exactly the four summands in (2).

$$\begin{array}{ccc} H_L^r(Y \times S_\sigma^1) & & r = 2n - k \\ \parallel & & \\ H^r(Y \times S_\sigma^1) \oplus H^{r-1}(Z \times S_\sigma^1) & & \\ \parallel & & \\ (H^r(Y) \oplus H^{r-1}(Y)) \oplus (H^{r-1}(Z) \oplus H^{r-2}(Z)) & & \end{array}$$

Figure 2: The four summands come from the ordinary class, the symbol-order logarithm $d \log r$, the boundary logarithm $d \log x$, and their product.

For a manifold with corners, the same local argument has one generator $d \log x_H$ for every boundary hypersurface H through a point. If $\mathcal{F}_j(M)$ denotes the codimension- j faces, then

$$H_L^r(M) \cong \bigoplus_{j \geq 0} \bigoplus_{F \in \mathcal{F}_j(M)} H^{r-j}(F; o(N_F)). \quad (8)$$

Here $o(N_F)$ is the orientation line of the conormal directions to F . If the hypersurfaces have been globally cooriented and ordered, these lines are trivialized and the untwisted formula results.

Indeed, apply the one-face residue successively. Residues for distinct hypersurfaces anticommute with the graded sign, so the summand attached to $F = H_1 \cap \cdots \cap H_j$ is represented by

$$\tilde{\gamma} \wedge d \log x_{H_1} \wedge \cdots \wedge d \log x_{H_j}, \quad [\gamma] \in H^{r-j}(F).$$

The product of the nonzero-weight contractions gives an acyclic complement. Changing the order of the hypersurfaces changes the orientation sign of the exterior logarithmic factors; the local system $o(N_F)$ records exactly this ambiguity. Iterated collar extensions provide a noncanonical splitting. This proves (8) rather than merely suggesting it from the one-boundary case.

8 Collapse and assembly of the computation

Combining HKR, the Poisson differential, symplectic duality, and homogeneity gives

$$E_k^2 \cong H_L^{2n-k}(\Upsilon \times S_\sigma^1).$$

The remaining input is the collapse theorem for Laurent complete symbols.

Theorem 8.1 (Benameur–Nistor collapse theorem; quoted). *Let G satisfy the anchor and local-triviality hypotheses of [1, equations (13)–(14)]. For its Benameur–Nistor topologically filtered Laurent complete-symbol algebra, the order spectral sequence converges and degenerates at E^2 . These hypotheses hold for the canonical b -groupoid $\mathcal{G}_b$ used in this manuscript.*

This is the global input from [2, 1]. It is stronger than a degree-counting assertion: higher differentials could a priori connect different symbol orders while preserving total degree. The theorem rules them out for the completed classical-symbol algebra.

The published collapse proof first treats a locally trivial product family, where the Künneth theorem produces Hochschild classes realizing every E^2 -class. Naturality then transfers survival to the completed product algebra, and multiplication by compactly supported base functions localizes the argument to trivializing sets. This proves vanishing of all higher differentials and the convergence bounds. It is not a “symplectic homotopy kills higher differentials” argument; symplectic duality computes E^2 , whereas locality and Künneth prove collapse.

Proposition 8.2 (Source dictionary). *The algebra used here is exactly the one-face b -groupoid specialization of the Benameur–Nistor theorem.*

Proof. The dictionary has six entries.

Groupoid. The differential groupoid is $\mathcal{G}_b$, with unit space X and Lie algebroid bTX .

Laurent multipliers. The coefficient ring is precisely $\mathcal{O}(X) = C^\infty(X)[x_H^{-1}]$.

Algebra and topology. The algebra is $\mathcal{O}(X)(\Psi^\infty(\mathcal{G}_b; E)/\Psi^{-\infty}(\mathcal{G}_b; E))$, equipped with the topologically filtered completion of [1]; its continuous Hochschild complex is the one used in their theorem.

Anchor assumption. With base reduced to a point, the fibre condition is automatic. The remaining assumption is exactly the Laurent anchor isomorphism proved in the anchor-bridge proposition.

Local triviality and coefficients. Benameur–Nistor prove that the anchor assumption supplies their required local topologically filtered models. Their global Morita lemma removes the coefficient bundle E .

Spectral sequence. Their filtered HKR proposition identifies E^1 ; their homogeneous Laurent-Poisson theorem gives E^2 ; and their collapse/convergence lemma identifies E^∞ with continuous Hochschild homology. Thus no independently asserted LF, patching, or Borel splitting hypothesis remains to be checked. $\square$

Proof of Theorem 1.1. Proposition 8.2 gives a convergent spectral sequence. The principal-symbol identification and HKR theorem give

$$E^1 \cong \Omega_L^*({}^bT^*X \setminus 0)_{\text{hom}},$$

and expansion (4) identifies d^1 with the Poisson boundary δ_b . Symplectic duality (6) reverses degree, and the Euler contraction removes every nonzero total weight. Therefore

$$E_k^2 \cong H_L^{2n-k}(Y \times S_\sigma^1).$$

The collapse theorem gives $E^2 = E^\infty$, so this is the associated graded of $\text{HH}_k^{\text{cont}}(\mathcal{A}_{b,L})$.

A Borel quantization splits the symbol filtration on the algebra, but it does not by itself split the induced filtration on homology. The Benamèur–Nistor theorem supplies the convergent filtration; over $\mathbb{C}$, choosing linear complements of its finitely many nonzero graded pieces gives the stated noncanonical vector-space isomorphism. The canonical assertion is the associated-graded identification (1). Finally, Proposition 7.1 and Künneth for S_σ^1 give the four summands in (2). $\square$

Discussion. Why is convergence not automatic? For a merely algebraic union of symbol orders, an infinite correction can escape the complex and an E^∞ -class need not lift to a chain. Borel summation and the strict Laurent inductive limit are ingredients in the topologically filtered axioms: the former realizes asymptotic symbols and the latter keeps every individual chain at finite pole order. They do not by themselves prove convergence. Convergence to Hochschild homology is the quoted Benamèur–Nistor spectral-sequence theorem, including its uniform vanishing-range verification.

9 A Čech–de Rham reconstruction

This section gives a second globalization of the filtered calculation. The essential preliminary step is localization of Hochschild chains to germs near the diagonal. Without that step, the assignment $U \mapsto C_q(C^\infty(U))$ is not a sheaf on one copy of the phase space: a degree- q chain naturally lives on the $(q+1)$ -fold product. After diagonal localization the chain germs do form a fine sheaf, and ordinary Čech descent becomes legitimate.

Put

$$M = {}^bT^*X \setminus 0, \quad Y = M/\mathbb{R}_+ = {}^bS^*X.$$

Let $\mathcal{F}_p C_q^{\text{cont}}$ be the total-symbol-order filtration defined in Section 2. For $p \in \mathbb{Z}$, depth $N \geq 0$, put

$$C_q^{p,N} = \mathcal{F}_p C_q^{\text{cont}} / \mathcal{F}_{p-N-1} C_q^{\text{cont}}.$$

This is a chain complex with the finite filtration range $p, p-1, \dots, p-N$, because the Hochschild boundary preserves total symbol order. Notice that we are quotienting the *chain filtration*; $F_m \mathcal{A}_{b,L}$ by itself is not an algebra for general m .

There is one choice to make before restricting these noncommutative chains. Fix a homogeneous complete-symbol splitting, compatibly on the cofinal Benamèur–Nistor stages. It represents $C_q^{p,N}$ by symbol jets on M^{q+1} with only $N+1$ retained total orders. Modulo the discarded orders, the Hochschild boundary is a finite sum of b-polydifferential operators followed by restriction to partial diagonals. Thus, for a conic open set $V \subset M^{q+1}$, restriction of the finite jets defines $C_q^{p,N}|_V$. A second symbol splitting differs from the first by an invertible triangular operator $\text{Id} + R$, where R lowers total order and is b-differential at each retained order. It therefore conjugates the two

restricted finite complexes. The localized complex below is independent of the splitting up to this filtered isomorphism.

Stagewise nuclearity identifies the associated graded tensors with smooth Laurent sections on M^{q+1} , with matrix coefficients suppressed by Morita invariance. The strict inductive limit over Laurent bounds is taken only after this construction on the cofinal stages; no fixed Laurent bound is incorrectly treated as a subcomplex. Let $\Delta_{q+1} \subset M^{q+1}$ be the total diagonal and define the localized finite strip by

$$C_q^{p,N,\text{loc}} = \varinjlim_{V \supset \Delta_{q+1}} C_q^{p,N}|_V, \quad (9)$$

where V runs through conic neighborhoods of the diagonal. The finite-jet formula above shows that the Hochschild boundary is local along the partial diagonals and therefore passes to the direct limit.

Proposition 9.1 (Filtered diagonal localization). *For every p and N , the germ map*

$$C_\bullet^{p,N} \longrightarrow C_\bullet^{p,N,\text{loc}}$$

is a filtration-preserving quasi-isomorphism. It is compatible with restriction to conic open sets, with enlargement of the Laurent bound, and with the truncation maps in N .

Proof. Filter both complexes by descending symbol order. Because the quotient has only the finite range $p, p-1, \dots, p-N$, this is a finite filtration. On the associated graded the product is commutative, and the chain complex is the completed smooth bar complex of the Laurent function algebra on M . The continuous bar-Koszul localization lemma used in the Brylinski-Getzler calculation [3] says that restriction to germs at the total diagonal is a quasi-isomorphism. Here is the local reason. Away from Δ_{q+1} , some two coordinate projections are separated. On a small product neighborhood choose a smooth function whose values on those two projections are disjoint. In the Koszul resolution of the diagonal, the corresponding difference is invertible and contraction by its inverse contracts the complex. A locally finite partition of unity on the complement of the diagonal patches these contractions. Multiplication by the cutoffs and all Koszul homotopies preserve finite Laurent pole order. Thus restriction to diagonal germs is a quasi-isomorphism on the associated graded complex.

The comparison theorem for finite filtered complexes now gives the result for $C_\bullet^{p,N}$. Equivalently, one may correct the commutative contraction successively: its commutator with the full Hochschild boundary lowers symbol order, and after at most N corrections the error vanishes in the quotient. Every correction is a finite sum of b-differential operations and smooth cutoffs, so it is continuous on each nuclear Fréchet stage and preserves finite Laurent bounds. The construction is local and hence commutes with restriction; it also commutes with the truncation and inductive-limit maps claimed in the statement. $\square$

Choose a finite good cover $\mathcal{U} = \{U_i\}$ of Y . Near $Z = \partial Y$, take collar products $[0, \varepsilon)_x \times V_i$, and shrink the cover so that a subordinate partition of unity has support compactly contained in the corresponding sets. Let $C_q^{p,N}$ be the sheaf on Y of conic diagonal germs from (9). If the first variable of a diagonal germ lies in the support of a partition function, then all variables lie in the same cover set after the diagonal neighborhood is chosen sufficiently small. Consequently multiplication in the first factor makes $C_q^{p,N}$ a fine C_Y^∞ -module. The cyclic term causes no support problem because all tensor variables are localized to the same diagonal germ.

Form the double complex

$$K_{p,N}^{a,q} = \check{C}^a(\mathcal{U}, \mathcal{C}_q^{p,N}), \quad D = \delta_{\check{C}} + (-1)^a b. \quad (10)$$

The total degree is $q - a$.

Lemma 9.2 (Čech descent after localization). *The augmentation*

$$C_{\bullet}^{p,N,\text{loc}} \longrightarrow \text{Tot } K_{p,N}^{\bullet,\bullet}$$

is a filtration-preserving quasi-isomorphism.

Proof. For fixed Hochschild degree, $\mathcal{C}_q^{p,N}$ is fine. The ordinary partition-of-unity homotopy contracts every augmented Čech row. Unlike the invalid contraction on unlocalized global chains, this homotopy lands in the stated local section because every tensor variable is a germ near the same diagonal point. It is continuous on each filtration stage and preserves Laurent and symbol bounds. Taking Čech cohomology first therefore leaves only degree zero, proving the assertion. $\square$

Filter (10) by total symbol order. On the associated graded, local filtered HKR and the local Koszul homotopy give

$$\check{C}^a(\mathcal{U}, \Omega_L^q(M)_{\text{hom}}). \quad (11)$$

The maps commute with restriction. The first order-lowering part of symbol composition gives $i^{-1}\delta_b$, exactly as in (5). Hence the next differential is the b-Poisson differential.

Proposition 9.3 (Čech computation of the second page). *For every p, N , the E^2 -page of (10) is the cohomology of the corresponding finite-order truncation of the homogeneous Laurent Poisson complex. These identifications commute with the truncation maps. In every stable bidegree, and hence after the finite strips exhaust the order filtration, the nonzero terms assemble in total degree k to*

$$H_L^{2n-k}(Y \times S_{\sigma}^1; \mathbb{C}).$$

The identification is independent of the good cover, partition of unity, and symbol splitting.

Proof. Symplectic duality (6) converts the Poisson boundary to the Laurent de Rham differential with complementary degree. The sheaves of Laurent forms with any fixed finite pole bound are fine, and multiplication by the partition of unity preserves that bound. Thus the Čech direction contributes no positive-degree cohomology and the hypercohomology is computed by global Laurent forms.

The Euler contraction removes every nonzero fiber weight. At weight zero a form is

$$\alpha_0 + \frac{dr}{r} \wedge \alpha_1,$$

so (7) identifies the remaining complex with Laurent forms on $Y \times S_{\sigma}^1$. A single finite strip can have artificial edge terms where the contracting homotopy leaves the strip; these disappear in every stable bidegree under enlargement. This is why the proposition asserts the geometric formula only for the compatible stable pages, not for an isolated truncation. Common refinements and the fine-sheaf homotopy show independence of the cover and partition. The triangular change-of-splitting operator constructed before (9) induces the identity on the associated graded, which proves independence of the symbol splitting and compatibility of the finite truncations. $\square$

The boundary contribution is read from the residue complex. One must not write the generally false sequence $0 \rightarrow \Omega^*(Y) \rightarrow \Omega_L^*(Y) \rightarrow \Omega^{*-1}(Z) \rightarrow 0$: a Laurent form with zero residue may still have higher poles. Put $\mathcal{K}^* = \ker(\text{Res}_Z)$. The nonzero normal-weight contraction from Proposition 7.1 gives a quasi-isomorphism

$$\Omega^*(Y) \longrightarrow \mathcal{K}^*.$$

The exact sequence

$$0 \longrightarrow \mathcal{K}^* \longrightarrow \Omega_L^*(Y) \xrightarrow{\text{Res}_Z} (\Omega^{*-1}(Z), -d_Z) \longrightarrow 0$$

therefore yields, after choosing a collar splitting,

$$R\Gamma(Y; \Omega_L^*) \simeq R\Gamma(Y; \Omega^*) \oplus R\Gamma(Z; \Omega^{*-1}). \quad (12)$$

This is the interior/boundary decomposition in (2); the residue filtration is canonical, while the displayed splitting is not.

Proposition 9.1 and Lemma 9.2 identify every finite filtered quotient of the original continuous chain complex with the corresponding Čech model. The Benaméur–Nistor topologically filtered theorem then supplies the uniform convergence and degeneration needed to pass through the completed classical filtration. Hence the Čech–de Rham construction is a genuine alternative globalization of the filtered proof, while the published theorem remains the input controlling the completed abutment.

10 A Rees–Fedosov reconstruction

The third route exposes the deformation behind symbol composition. There are again two logically distinct tasks. First, the actual Rees symbol product is represented by a Fedosov connection with its own characteristic class, and Lie-algebroid chain formality computes the corresponding formal local model. Second, finite Rees jets are compared with finite classical symbol jets. The Benaméur–Nistor theorem is still used for convergence of the completed classical filtration.

10.1 The actual Rees deformation

The algebraic Rees algebra is

$$\mathcal{R}_F(\mathcal{A}_{b,L}) = \bigoplus_{m \in \mathbb{Z}} t^m F_m \mathcal{A}_{b,L} \subset \mathcal{A}_{b,L}[t, t^{-1}].$$

For a fixed upper-order sector, factor out t^m and encode

$$a \sim a_m + a_{m-1} + \cdots$$

by

$$a_m + \hbar a_{m-1} + \hbar^2 a_{m-2} + \cdots. \quad (13)$$

Here t^m records absolute order and $\hbar$ records loss of order; (13) is an adic chart on one sector, not a global $\mathbb{C}[[\hbar]]$ -module structure on the union of all orders.

Full symbol composition becomes

$$a \star c = \sum_{j \geq 0} \hbar^j C_j(a, c), \quad C_0(a, c) = ac, \quad (14)$$

where every C_j is b-bidifferential and

$$C_1(a, c) - C_1(c, a) = i^{-1}\{a, c\}_b.$$

Thus this is a differential deformation of the symplectic Lie algebroid $({}^bTM, \omega_b)$.

Theorem 10.1 (Fedosov representation of the symbol deformation; quoted). *The formal Rees product (14) is isomorphic, by an operator $T = \text{Id} + O(\hbar)$, to the flat-section product of a Fedosov connection on the b-Weyl bundle whose characteristic class is the characteristic class of the Rees deformation. The equivalence T and its inverse may be chosen coefficientwise b-differential; consequently they preserve the strict union of finite Laurent stages.*

This is the Fedosov-representation and classification theorem for symplectic Lie algebroid deformations of Nest–Tsygan [9]. The point is not that an arbitrary Fedosov product with the same first Poisson bracket is equivalent to the symbol product; one must choose the Fedosov curvature representing the same deformation class. With that choice, equivalent products have isomorphic Hochschild complexes. Moreover, the equivalences in the Lie-algebroid classification are Lie-algebroid differential operators. For bTM , their coefficients are smooth in the b-structure, so b-normal and tangential derivatives preserve finite Laurent pole order. This proves the Laurent-preservation clause in the quoted theorem and is essential when passing from the Rees product to its Fedosov representative.

10.2 Laurent local chain formality

Let $\mathcal{C}_{\bullet}^{\text{loc}, \text{diff}}$ denote the sheaf of localized differential Hochschild chains of the formal Laurent star algebra. The localization is the formal analogue of Proposition 9.1; it is the natural chain object for the Kontsevich–Shoikhet and Fedosov resolutions.

Proposition 10.2 (Laurent Lie-algebroid chain formality). *For the Fedosov representative in Theorem 10.1, there is a quasi-isomorphism of sheaves of $\mathbb{C}((\hbar))$ -complexes*

$$\left(\mathcal{C}_{\bullet}^{\text{loc}, \text{diff}}, b_{\star}\right) \simeq \left(\Omega_L^{\bullet}(M)((\hbar)), \mathcal{L}_{\pi_{\hbar}}\right), \quad (15)$$

where $\pi_{\hbar} = \hbar\pi_b + O(\hbar^2)$ is the formal Poisson tensor. Here $\mathcal{L}_{\pi_{\hbar}} = [\iota_{\pi_{\hbar}}, d]$. The maps and homotopies preserve finite Laurent pole order coefficientwise in $\hbar$.

Proof. Calaque–Dolgushev–Halbout prove Hochschild-chain formality for Lie algebroids by a Fedosov resolution and local Kontsevich–Shoikhet maps [4]; Dolgushev’s theorem is the smooth tangent-bundle prototype [5]. Apply their construction to the b-tangent Lie algebroid and twist by the Maurer–Cartan element of the Fedosov representative of the actual Rees product.

It remains to check the Laurent restriction. In a corner chart the local Taylor components, Fedosov differential, twisting operators, and homotopies are universal polydifferential expressions in b-vector fields and in coefficients smooth in the b-structure. A b-normal derivative satisfies $x_H \partial_{x_H}(x_H^{-\ell}) = -\ell x_H^{-\ell-1}$, while tangential b-derivatives preserve the pole order. Hence, in each Hochschild degree, every coefficient of every formality map sends a fixed finite Laurent stage into another finite stage. The Lie-algebroid Fedosov globalization is intrinsic on the b-tangent algebroid and makes these local maps compatible on overlaps. Passing through the strict union of pole bounds gives (15), and the same argument applies to its homotopies. $\square$

The formal tensor $\pi_{\hbar}$ is symplectic over $\mathbb{C}((\hbar))$. If $\omega_{\hbar}$ is its inverse, formal symplectic duality converts $\mathcal{L}_{\pi_{\hbar}}$ to the de Rham differential, up to the usual degree sign and an invertible scalar. Since the Laurent-form sheaves are fine, global sections compute their hypercohomology. Therefore

$$\mathrm{HH}_k^{\mathrm{loc}, \mathrm{diff}, \mathbb{C}((\hbar))}(\mathcal{O}_L(M)((\hbar)), \star) \cong H_L^{2n-k}(M; \mathbb{C})((\hbar)). \quad (16)$$

Equation (16) computes the localized differential/continuous formal complex of Proposition 10.2. It is not an unstated identification with unrestricted algebraic Hochschild homology. It has no independent order-circle class, because the complex is relative to $\mathbb{C}((\hbar))$ and hence $d\hbar = 0$. The unique S_{σ}^1 class arises later from the radial weight-zero form dr/r in the classical homogeneous order complex. We deliberately do not assume that a chosen formality quasi-isomorphism is equivariant for positive fiber dilations; no such equivariance is needed below.

10.3 From formal series to classical complete symbols

Formal formality does not by itself compute continuous homology of $\mathcal{A}_{b,L}$. The bridge to the classical filtration is finite-order symbol calculus.

Proposition 10.3 (Finite-order Rees comparison). *After fixing a homogeneous classical symbol map, for every $m \in \mathbb{Z}$ and $N \geq 0$, the order- m symbol-jet space has a continuous linear identification*

$$J_m^N := F_m \mathcal{A}_{b,L} / F_{m-N-1} \mathcal{A}_{b,L} \cong \bigoplus_{j=0}^N \mathcal{O}_L(M)_{m-j}. \quad (17)$$

These spaces are not asserted to be algebras. For every pair m, m' , composition induces

$$J_m^N \times J_{m'}^N \longrightarrow J_{m+m'}^N,$$

and under (17) this bilinear map is (14) modulo $\hbar^{N+1}$. The identifications are compatible with truncation and, after summing the orders of all tensor factors, with the finite quotients of the total-order Hochschild filtration. Consequently the Rees and classical order spectral sequences have the same E^1 -page and Poisson differential d^1 .

Proof. At finite N , Taylor expansion in symbol order gives (17) without an infinite summation. The b-symbol composition formula identifies the products coefficient by coefficient modulo $\hbar^{N+1}$, and its first antisymmetric correction is the b-Poisson bracket.

For a fixed Laurent bound, both sides are finite direct sums of nuclear Fréchet spaces. The coefficient and truncation maps are continuous. For a chain, apply these coefficient maps factor by factor and retain the terms whose total order lies in the chosen finite strip. Taking completed projective tensor powers first and the strict inductive limit over the Benaméur–Nistor stages afterward is exactly the order used in (3); hence the comparison extends to the finite quotients of the total-order continuous Hochschild filtration.

Finally, the b-symbol Borel lemma realizes every compatible homogeneous sequence by a classical polyhomogeneous symbol, uniquely modulo smoothing. This identifies the projective limit of algebraic symbol truncations with complete symbols. It does not assert that Hochschild homology commutes with the projective/inductive limits; that convergence statement remains the Benaméur–Nistor input. $\square$

The finite comparison identifies the classical E^1 -page and $d^1 = \delta_b$. Symplectic duality and the Euler contraction then give

$$E_k^2 \cong H_L^{2n-k}(Y \times S_\sigma^1; \mathbb{C}).$$

The formal quasi-isomorphism (15) explains why the local deformation has a de Rham model to all formal orders, but the homogeneous weight reduction is performed on the classical associated graded and does not rely on an unproved equivariant formality theorem. Benamèur–Nistor convergence and degeneration finally identify this second page with the associated graded of continuous Hochschild homology and give the noncanonical vector-space splitting.

Theorem 10.4 (Rees–Fedosov reconstruction). *The actual Rees symbol deformation has localized formal Hochschild homology (16). Its finite jets agree with the finite classical symbol quotients by Proposition 10.3; the classical homogeneity calculation supplies the single order circle. Together with the Benamèur–Nistor convergence-and-collapse theorem, these facts recover the associated-graded and noncanonical vector-space assertions of Theorem 1.1.*

Thus the Rees–Fedosov route is not a forbidden specialization of formal homology at $\hbar = 1$. Formality computes the localized formal model, finite Rees jets compare it with the classical filtration, and the published topologically filtered theorem controls the completed limit.

$$\begin{array}{ccc}
C_{\bullet}^{\text{cont}}(\mathcal{A}_{b,L}) & \xrightarrow{\text{diagonal localization}/\check{\text{Cech}}} & \text{Tot } \check{C}^{\bullet}(\mathcal{U}, \mathcal{C}_{\bullet}^{\text{loc}}) \xrightarrow{\text{HKR}+\delta_b} \Omega_L^{2n-\bullet}(M)_0 \\
\downarrow \text{compatible finite quotients} & & \nearrow *_{\omega_{\hbar}} \\
C_{\bullet}^{\text{loc,diff}}(\mathcal{O}_L(M)((\hbar)), \star) & \xrightarrow{\text{Fedosov formality}} & (\Omega_L^{\bullet}(M)((\hbar)), \mathcal{L}_{\pi_{\hbar}})
\end{array}$$

Figure 3: The Čech route first localizes chains to the diagonal and then globalizes the filtered HKR–Poisson calculation. The Rees route represents the actual symbol deformation by a Fedosov connection and computes its localized formal Hochschild complex. Finite Rees jets connect the formal and classical filtrations; Benamèur–Nistor controls the completed abutment.

11 Checks and consequences

11.1 The closed-manifold case

If $\partial X = \emptyset$, Laurent cohomology is ordinary cohomology and $Z = \emptyset$. Formula (2) reduces to

$$\text{HH}_k^{\text{cont}}(\mathcal{A}(X; E)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X),$$

equivalently $H^{2n-k}(S^*X \times S_\sigma^1)$. This is the Brylinski–Getzler computation [3].

11.2 Degree zero and residue traces

For the completed chain complex (3), degree-zero Hochschild homology is

$$\text{HH}_0^{\text{cont}}(\mathcal{A}) = \mathcal{A} / \text{im}(b : C_1^{\text{cont}} \rightarrow \mathcal{A}).$$

We abbreviate the image by $[\mathcal{A}, \mathcal{A}]_{\text{cont}}$; it need not be closed. Its Hausdorffization is the different quotient

$$\text{HH}_0^{\text{cont}}(\mathcal{A})_{\text{H}} = \mathcal{A} / \overline{[\mathcal{A}, \mathcal{A}]_{\text{cont}}}.$$

Theorem 1.1 computes the former group (and its filtered associated graded); it does not prove that the commutator image is closed. Continuous traces factor through the Hausdorffization.

For the canonical b -groupoid, the continuous-trace theorem of [1] identifies those traces with the top Laurent residue classes. If $n \geq 2$, X is connected, and ∂X is connected, the boundary cosphere Z is connected and carries its canonical contact/boundary orientation, so $H^{2n-2}(Z) \cong \mathbb{C}$. The trace theorem, not an unproved closedness claim, therefore shows that the space of continuous traces is one-dimensional.

If $n \geq 2$, X is connected, and ∂X is connected, then the resulting trace is the b -residue trace up to scale. With several boundary components, the same theorem gives one boundary-face residue for each connected component of Z .

11.3 A product cylinder

Let $X = [0, 1] \times N$, where N is closed of dimension $n - 1$, and put

$$W = S(\mathbb{R} \oplus T^*N).$$

Then $Y \simeq [0, 1] \times W$ and $Z = W \sqcup W$. Therefore

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-k}(W) \oplus H^{2n-k-1}(W)^{\oplus 3} \oplus H^{2n-k-2}(W)^{\oplus 2}.$$

This example separates the order-circle contribution from the two boundary faces.

12 The unreduced operator algebra

The unreduced operator algebra fits into

$$0 \longrightarrow \Psi_{b,L}^{-\infty}(X; E) \longrightarrow \Psi_{b,L}^{\mathbb{Z}}(X; E) \longrightarrow \mathcal{A}_{b,L}(X; E) \longrightarrow 0. \quad (18)$$

The answer for $\Psi_{b,L}^{\mathbb{Z}}$ is not obtained by simply reusing (1). It depends on the topology placed on the residual ideal and on whether kernels are required to vanish rapidly at the side faces, merely have polyhomogeneous expansions, or are completed in an operator norm.

For an H -unital residual ideal, excision gives the long exact sequence

$$\cdots \rightarrow \mathrm{HH}_k(\Psi_{b,L}^{-\infty}) \rightarrow \mathrm{HH}_k(\Psi_{b,L}^{\mathbb{Z}}) \rightarrow \mathrm{HH}_k(\mathcal{A}_{b,L}) \xrightarrow{\partial} \mathrm{HH}_{k-1}(\Psi_{b,L}^{-\infty}) \rightarrow \cdots$$

Melrose and Nistor compute the analogous two-filtration sequence in the cusp calculus and explain how the same mechanism applies to the b -calculus [8].

Here H -unitality means that the completed bar complex of the nonunital residual ideal is acyclic; this is the hypothesis that prevents an additional excision defect. Continuous excision and H -unitality are substantial theorems of Wodzicki type [12]. For the standard rapidly decreasing nuclear smoothing algebra the hypothesis is known, but it is not proved by a one-line finite-rank approximation. Allowing polyhomogeneous side-face terms changes both topology and residual homology, so H -unitality must be verified anew for the chosen completion.

The Hochschild connecting morphism can be seen on a symbol cycle. Choose operator lifts of its tensor factors, apply the Hochschild boundary upstairs, and observe that the result lies in the residual ideal because its image vanishes in the quotient. The resulting residual homology class is independent of the lifts. This Hochschild map is not literally the Fredholm boundary map in K -theory. Compatibility appears only after applying the Chern character to cyclic homology and

using excision in both theories. Boundary logarithmic classes then pair with regularized cyclic cocycles; the eta term is an analytic realization of such a pairing, not an element produced by the Hochschild boundary alone.

Thus (1) is the complete answer for Laurent complete symbols, while (18), together with the homology of the chosen residual ideal, is the correct computation for a specified unreduced operator algebra.

13 Relation to index theory

An elliptic b-operator determines an invertible matrix over the complete-symbol algebra. A Fredholm index additionally requires invertibility of its normal family. Its K_1 -class has a Chern character in cyclic homology, which can pair with cyclic cocycles arising from the Hochschild classes above. In this qualified sense, the ordinary residue class supplies the interior symbolic term and a boundary logarithmic cocycle supplies the indicial correction. The operator-extension boundary map is the Fredholm index only under full ellipticity.

The work of Liu and Ma on differential K -theory and eta-invariant localization supplies a geometric refinement of this pairing: differential forms and eta data refine the topological K -class. It does not replace the Hochschild spectral-sequence calculation [6]. The logical division is therefore

Melrose–Nistor and Benameur–Nistor: algebraic homology,

Liu–Ma: differential- K and eta localization of index data.

Mazzeo–Melrose’s fibred-boundary calculus gives complementary analytic background for normal families, mapping properties, and Fredholm criteria [10]. It does not replace the Benameur–Nistor topologically filtered Hochschild theorem. This distinction is useful when transporting the present computation to edge or fibred-cusp settings.

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